FIGARO. PROTOCOL

The TCP/IP of Trade — self-enforcing agreements between strangers.

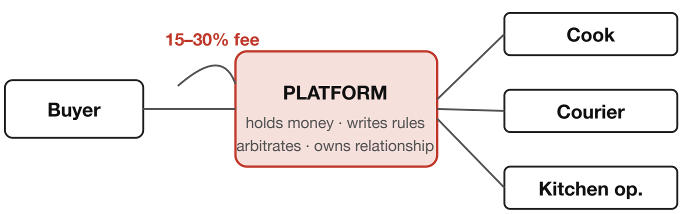
Figaro is a **stateless, ownerless coordination protocol**. It defines the smallest possible unit of a secure handshake — the ***bonded commitment*** — so that two people who have never met can transact with mathematical certainty that cooperation is the dominant strategy. No arbitrator. No timeout. No admin backdoor. **Not DeFi. Not TradFi.** A primitive, like TCP/IP, that anything else can be composed on top of.

THE PARADIGM SHIFT

TODAY — PLATFORM MODEL	FIGARO — BONDED MODEL
Trust is rented from a company. A standing firm (Uber, DoorDash, Amazon) sits between every buyer and every worker. It holds the money, writes the rules, arbitrates disputes, controls the relationship, and takes 15–30% as rent for providing trust.	**Trust is produced by the deal itself.** Both parties lock **2× collateral**. Cheating always costs more than cooperating, so neither party defects. Each transaction assembles its own temporary institution from the people actually doing the work — then dissolves at settlement.

TODAY

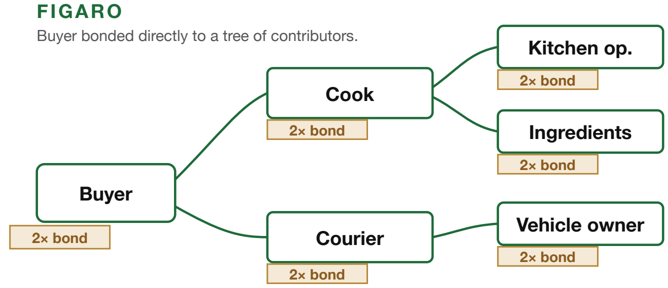
Buyer and workers routed through a firm.



workers are labor; the firm is the institution

FIGARO

Buyer bonded directly to a tree of contributors.



transaction-scoped institution — assembled at commit, dissolved at settlement

if anyone fails, everyone loses their bond → peers police each other (micro-lending-circle effect)

HOW THE HANDSHAKE WORKS

1 BOND

Buyer and seller dual-sign an EIP-712 commitment. Both bonds pull atomically: buyer posts **2× payment**, seller posts **2× cumulative value**. At 2× the ratio, cheating is always a net loss — the minimum that makes every form of defection irrational (Nash).

2 PERFORM

Work is performed off-chain. Each lifecycle event (prepared, picked up, delivered) is written as a **role-gated, timestamped attestation** on the blockchain — tamper-proof evidence produced as a side effect of the mechanism, not as a ceremony.

3 RESOLVE

The buyer triggers one **atomic resolution**. Bonds return to every contributor; payment settles. The entire process tree clears in a single transaction. No arbitrator decides. No timeout releases funds. No admin key exists.

THREE LAYERS OF ENFORCEMENT

ECONOMIC

Deposits make cooperation the dominant strategy. Covers **>99%** of transactions — rational actors never defect.

SOCIAL

Multi-party trees bind contributors: one failure collapses the whole settlement, so everyone polices everyone else — the same mechanism that drops micro-lending defaults from ~20% to ~2%.

LEGAL

For the irrational fringe, the chain has already recorded immutable, timestamped evidence. Courts don't reconstruct what happened — it's on-chain.

WHAT IT UNLOCKS

Any scenario where strangers need to coordinate reliably: **delivery** (food, parcels, groceries) · **ride-hail** (driver, vehicle owner, passenger) · **multi-tier procurement** · **freelance** milestone work · **diaspora and community networks** · **AI-agent-to-AI-agent commerce**. The protocol is permissionless, token-agnostic (any ERC-20), open source, and works identically in New York, Nairobi, or anywhere an internet connection reaches. *Figaro Eats* is the first working archetype — a complete food-delivery institution composed from the permissionless primitives.

Status — Canonical runtime: `Figaro-Prototype2`. Solidity surface frozen for external audit. Kernel = 2 external functions, 3 mappings, no owner. 227 Foundry tests, 15 TLA+ invariants, 11 Halmos symbolic proofs, 27 Certora rules, Echidna fuzzing.

Read next: VISION.md · THEORY.md · RUNTIME_THESIS.md